Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/13/2007
PI Name:	Leal, L. Gary Ph.D.		
Project Title:	Flow-induced Coalescence of Drops		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	patk@engineering.ucsb.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	805-893-8510
Organization Name:	University of California - Santa Barbara		
PI Address 1:	Department of Chemical Engineering		
PI Address 2:	Rm. 3353, Engineering II		
PI Web Page:			
City:	Santa Barbara	**State:**	CA
Zip Code:	93106-5080	**Congressional District:**	23
Comments:	PI email: lgl20@engineering.ucsb.edu Co-I email: jacob@engineering.ucsb.edu PI: URL: http://www.engineering.ucsb.edu/		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Sankaran, Subramanian	**Contact Phone:**	216-433-9335
Contact Email:	Sankaran@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04GA58G		
Performance Goal No.:			
Performance Goal Text:	NOTE: Original end date was 11/30/2007		

Task Description:	We propose a fundamental study of flow-induced coalescence in viscous fluids. This project will be based upon both macroscale experiments involving the collision and possible coalescence of two drops (1-100 microns in diameter) in a flow, and sub-microscale experiments providing details of the film drainage process, including the shape of the deforming film and the molecular-scale process that leads to rupture and coalescence. The macroscale experiments will use the 4-roll mill to extend current studies to non-dilute systems, where the crowding with neighboring drops will affect the collision and coalescence process, as well as a new cross-slot flow device to study drops in the size range from 1-10 microns in diameter. The micro- and nano-scale studies will be carried out in a surface forces apparatus (SFA) that has been specifically modified for this project. We will study systems both with and without surfactant. The immediate objectives of this study are to determine the mechanisms for coalescence at a detailed level so that we can understand recently obtained macroscale results, such as the observation that coalescence often occurs after two drops have collided and are being pulled apart by the external flow. We also intend to extend the existing macroscale studies to drop sizes and concentrations that are relevant to many applications. The coalescence of drops in a viscous fluid is a key component in understanding and controlling the drop size distribution in polymer blends, where the surfactant is a copolymer, known as a compatibilizer. Reactive blending is the most important source of new polymeric materials. The drop size and interface properties are the key to determining rheological and other properties of both blends and other fluid-fluid colloidal dispersions. Flow-induced coalescence is also an important phenomenon during the separation of fluids or materials in microgravity. Finally, the eventual possibility of doing microgravity coalescence experiments is very important in extending this work to systems such as gas-liquid dispersions, where there is a significant density difference between the two fluids. The 4-roll mill is a small compact device that could be flown in microgravity. It can be operated completely remotely, and could actually be a very significant enhancement of microgravity flow facilities for a variety of fluids and materials applications (which may work at the Glove-box level). The objectives of this project are two-fold: 1) To study the coalescence process in the Surface Forces Apparatus under conditions relevant to the flow-induced coalescence process for small drops in a viscous liquid. The objective of this work is to allow a direct detailed investigation of the film thinning and rupture process using an axisymmetric SFA configuration, and 2) To focus the 4-roll mill experiments on smaller drops, which are more in the range of direct technological significance, and which will also allow us to achieve a better understanding of the coalescence process. One of the important issues associated with long-range flight, exploration, and development in space is the management of multiphase fluid systems. Phase separation can be achieved via thermocapillary-driven coalescence, and diffusion driven phenomena; however, in many cases, coalescence driven directly by velocity gradients of a flow will be the dominant and most viable mechanism. The deformation, breakup, and coalescence dynamics of drops and bubbles are of fundamental concern in many 1-g processes, such as polymer blending, mixing, emulsification etc, but are not well understood at a fundamental level. The absence of buoyancy- driven coalescence and the very slow time scales of thermocapillary driven processes suggests that there will be many examples of multiphase flow processes where the processes of coalescence or breakup driven by velocity gradients of a mean flow may be of especially great importance for 0-g applications. The four-roll mill generates ideal flow fields to study the dynamics of neutrally buoyant particles or cluster of particles. Both flow-driven breakup and coalescence processes can be studied in transient and steady flows ranging from purely extensional to simple shear with the positions of the drops controlled by a real time imaging/computer control algorithm. The ground based work in this project will pave the way to microgravity studies on non-neutrally buoyant systems(e.g., bubbles). Some preliminary ground-based work is possible using acoustic centering systems to counteract the effects of gravity. Such a ‘flow facility’, when operated in microgravity, will facilitate studies on non-neutrally buoyant systems in a variety of steady and time-dependent flows, and may be applicable to many systems beyond the coalescence studies of this project. The Surface Forces Apparatus (SFA) will allow us to measure various surface and thin film shapes and forces for normal and tangential motions of bubbles and drops. The key objective of these experiments is the determination of the non-hydrodynamic forces that operate between pairs of liquids, and which are responsible for the instability that leads to rupture of the thin film in coalescence processes. Such micro scale experimental results would be tied to the macro scale experimental results from the four-roll mill experiments via numerical modeling and computations. This approach will 1) clarify the nature of the hydrophobic interactions between pairs of fluids, 2) establish the limits of applicability of macroscopic concepts, such as contact-angle, at the sub-microscopic and molecular levels; thus, a more elegant and more realistic understanding of the bubble and drops interaction, collision, coalescence, and break-up processes is achievable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The deformation, breakup, and coalescence dynamics of drops and bubbles are of fundamental concern in many 1-g processes, such as polymer blending, mixing, emulsification, responses of biological cells, etc, but are not well understood at a fundamental level. The results from this study will be of interest to a variety of such industries.
	The objectives of the modified work statement that was submitted at the point of funding this project were two-fold: 1) To study the coalescence process in the Surface Forces Apparatus under conditions relevant to the flow-induced coalescence process for small drops in a viscous liquid. The objective of this work is to allow a direct detailed investigation of the film thinning and rupture process using an axisymmetric SFA configuration. and 2) To focus the 4-roll mill experiments on smaller drops, which are more in the range of direct technological significance, and which will also allow us to achieve a better understanding of the coalescence process. In addition, we proposed to continue work via the SFA apparatus on the origin of hydrophobic forces and on coalescence with extremely weak (i.e. Ca~0) “flow” conditions. We have, in fact, made progress on all of these fronts, and in addition have incorporated theoretical work that explores the dynamics and stability of the thin films that arise in the 4-roll mill coalescence experiments, but cannot be seen experimentally. As far as objective #1 is concerned, we have now built an SFA attachment that allows film drainage studies in both axisymmetric and cross-cylinder geometries, under constant force conditions, and initial experiments are underway to test this fundamentally new system. Initial experiments were carried out where the approach force was constant rather than the approach speed, using PS 800 films of thickness T~200 nm on mica, with the surfaces were totally immersed in

Task Progress:

PDMS (Mw= 6,000).

Figure 1. FECO fringes of two approaching surfaces supported by the new “balance attachment” where coalescence occurs at the rim of the contact circle.

The results may be directly relevant or comparable to the conditions for surface deformation, and film thinning and rupture in four-roll mill flow-induced coalescence experiments. We are currently analyzing the results of experiments that give the surface shape profiles (i.e., the surface separation profiles) as a function of time for different forces and for liquids of different viscosities. We will also use the high-speed camera, as required, to image the rapid transient effects occurring during coalescence (described above); and we are also setting up a the new high-resolution CCD camera data acquisition system with its image processing software for this project. Our experiments using the SFA and new balance attachment corroborate the theoretical results described below in showing that there is more than one route to coalescence, the differences being reflected most dramatically in the different deformations of the surfaces leading up to coalescence. The mechanism or process that actually occurs depends on the nature of the attractive (adhesion) forces between the two liquid surfaces across the host liquid, the viscosity ratios of the two liquids, the capillary number based on the approach velocity V (or an equivalent measure of the force in constant force experiments), the geometry of the coalescing surfaces, and the film thickness T (relative to the radius R). The experiments, however, also allow us to examine the surface deformations during and after coalescence. The two main types of micron- or submicron-sized deformations (i.e., much smaller than the contact radius r) that we observe on coalescence or detachment are ripples and fingering. Our analysis of ripples (by Howard Stone) shows them to be of the right wavelength and expected given the van der Waals forces between the surfaces, the gap width at that point, the interfacial tension, and viscosity. In the case of fingering the theoretical maximum wavelength of fingers for bulk systems is given by the equation: , where V is the velocity of the interface in the direction from liquid 1 to liquid 2, ?i the viscosities of the media, ? the interfacial tension, and b the slit thickness. Our limited results and analysis to date confirm that the above equation holds even for micron sized fingers. Further, while fingering always occurs for straight interfaces (if), for curved (circular) bridges or necks of radius r fingering instabilities occur only if This is consistent with the lack of any reports (by us or others) of fingering instabilities in narrow low viscosity liquid bridges as they thin.

We have also rebuilt one of our two 4-roll mills to allow experiments to be carried out for much smaller drop sizes (2-3 micron drops). This is critical to understanding the coalescence process under circumstances where the van der Waals attraction influences not only the film-drainage and rupture process but is also important in determining the overall collision trajectories. It is also critical to assessing the role of surfactant because changes in drop size allow “independent” variation in the capillary number and the interface Peclet number. During the past year, experiments to utilize this new facility have been carried out, and we are still in the process of doing additional experiments and analyzing the data that we have already obtained, with a major focus on the role of surfactant on drop deformation and breakup, as well as on coalescence. Finally, we have initiated some experiments on coalescence between drops in more concentrated systems, by simulating the effects of crowding using refractive index matched particles.

We have also made considerable progress in theoretical modeling of the coalescence process, and especially the conditions leading to film rupture. This includes (1) asymptotic theory based on the thin-film approximation (during the past year this work has been prepared and accepted for publication); (2) direct numerical solution of the two-drop collision and coalescence process using boundary integral methods (again, though others have also pursued this approach, we have shown that a much greater degree of resolution is required in the thin film region than is inherent in other attempts at numerical simulation (at least for the extremely small capillary numbers that characterize the coalescence process in our experiments), and this has required the development of grid refinement techniques to obtain sufficiently accurate solutions). During the past year, the source of an apparent discrepancy between the expected theoretical value of the Hamaker constant, and the value that seemed to provide the best theoretical predictions was identified, and a very extensive series of calculations has been carried out for two drops without surfactant. These show good qualitative agreement with experimental data, but there are discrepancies in detailed predictions and some phenomena that are seen experimentally but not predicted theoretically. We are currently seeking the source of these differences, including the possibility that the film thickness is too small to allow continuum theory to be accurate; (3) linear stability theory to examine the role of thermal fluctuations and non-axisymmetric perturbations of shape on the conditions for film rupture. The distinction here with the many past studies of thin film stability is that we account for the time-dependent evolution of the film shape, and for the finite and evolving radial extent of the thin film. We have continued to work on this theory to account for interface mobility and the actual shapes of the thin films.

Finally, we have continued our work on hydrophobic force measurements. The hydrophobic interaction is among the most important non-specific interaction in biological and many colloidal systems, for example, between oil and emulsion droplets in water. Our SFA studies on the elucidation of this interaction continues. We have published two papers on this in 2005 (described below), and are continuing with our attempt to measure the short-range force function in the short-distance regime from distances (surface separations) of D=5 nm down to contact (D=0) where the thermodynamic adhesion force or energy of ?~50 mJ/m2 should be measured on separation, as well as the deformations accompanying the adhesion, coalescence and detachments of droplets in the SFA. As studies of the hydrophobic interaction continue, it has become increasingly clear that the long-range attraction observed in many experiments is not in fact due to any direct hydrophobic attraction at all. Our current aim is to understand the ‘real’ hydrophobic interaction between molecularly smooth, hydrophobic layers that are chemisorbed (rather than physisorbed surfactant monolayers, which are unstable in water, exhibit contact angle hysteresis, and do not give the correct surface energy when the surfaces are in molecular contact). We have measured the attractive long-ranged ‘hydrophobic’ forces in water between double-chained surfactant monolayers physisorbed on mica. We used both normal and high-speed video cameras to follow the dynamics of the interaction forces (time-dependence of the force-distance profiles) in the distance regime from D=1,000Å to adhesive contact (D=0), concentrating on the short-distance regime below 200Å – the regime of greatest colloidal and biological interest. Our results show that the attractive force-law changes from exponential at large separations to an inverse power-law, of the type F?-1/D between two curved surfaces, at separations from 200Å down to adhesive contact under the assumption of a no-slip boundary condition at each surface (zero slip length). However, the data can also be fitted assuming a different short-range force and a finite slip-length of 150-200 Å, which would be consistent with recent studies by others using different techniques. The results of these SFA experiments showed that dissolved gases have a strong effect on the hydrophobic interaction. However, unlike what was previously thought, they are not the cause of this interaction, i.e., pre-existing air bubbles or cavities on isolated hydrophobic surfaces do not exist nor are they responsible for the strong attraction between hydrophobic surfaces or oil droplets in water. However, such bubbles or cavities may be nucleated?? between two approaching hydrophobic surfaces, pulling them into contact via the capillary force due to the Laplace pressure, but only when the surfaces are very close together (D<1-2 nm) and only when the

	contact angle is greater than 90°.
Bibliography Type:	Description: (Last Updated: 03/13/2007)
Articles in Peer-reviewed Journals	Baldessari F, Leal LG. "Two touching spherical drops in uniaxial extensional flow: Analytic solutions to the creeping flow problem." J Colloid Interface Sci. 2005 Sep 1;289(1):262-70. PMID: 16009232 , Sep-2005
Articles in Peer-reviewed Journals	Lin Q, Meyer EE, Tadmor M, Israelachvili JN, Kuhl TL. "Measurement of the Long- and Short-Range Hydrophobic Attraction between Surfactant-Coated Surfaces" Langmuir 2005 Jan 4;21(1):251-5. PMID: 15620310 http://dx.doi.org/10.1021/la048317q , Jan-2005
Articles in Peer-reviewed Journals	Meyer EE, Lin Q, Israelachvili JN. "Effects of dissolved gas on the hydrophobic attraction between surfactant-coated surfaces." Langmuir 2005 Jan 4;21(1):256-9. PMID: 15620311 http://dx.doi.org/10.1021/la048318i , Jan-2005
Articles in Peer-reviewed Journals	Yoon Y, Borrell M, Park CC, Leal LG. "Viscosity ratio effects on the coalescence of two equal-sized drops in a two-dimensional linear flow." Journal of Fluid Mechanics 2005 Feb;525:355-79. http://dx.doi.org/10.1017/S0022112004002824 , Feb-2005
Articles in Peer-reviewed Journals	Zeng H, Maeda N, Chen N, Tirrell M, Israelachvili J. "Adhesion and friction of polystyrene surfaces around Tg." Macromolecules 2006 Mar 21;39(6):2350-63. http://dx.doi.org/10.1021/ma052207o , Mar-2006
Articles in Peer-reviewed Journals	Meyer EE, Rosenberg KJ, Israelachvili J. "Recent progress in understanding hydrophobic interactions." Proc Natl Acad Sci U S A. 2006 Oct 24;103(43):15739-46. PMID: 17023540 , Oct-2006
Articles in Peer-reviewed Journals	Zeng H, Zhao B, Tian Y, Tirrell M, Leal LG, Israelachvili J. "Transient surface patterns during adhesion and coalescence of thin liquid films." Soft Matter 2007;3:88. http://dx.doi.org/10.1039/b613198k , Jan-2007
Papers from Meeting Proceedings	Leal LG. "Studies of Droplet Coalescence and Breakup with Application to Polymer Blending." 4th Pacific Rim Conference on Rheology, 2005. Advances in Rheology and Its Applications, Proceedings of 4th Pacific Rim Conference on Rheology, pp. 1-4, 2005. , Jan-2005